A Technical Introduction to Adversarial machine learning

Section 1

Repeat this for v 2 , v 3 , … {\displaystyle v_{2},v_{3},\dots } until the desired level of error in the classifier is reached.It was discovered when the authors designed a simple baseline to compare with a previous black-box adversarial attack algorithm based on gaussian processes, and were surprised that the baseline worked even better.

Section 2

Black box attacks Black box attacks in adversarial machine learning assume that the adversary can only get outputs for provided inputs and has no knowledge of the model structure or parameters. In this case, the adversarial example is generated either using a model created from scratch, or without any model at all (excluding the ability to query the original model). In either case, the objective of these attacks is to create adversarial examples that are able to transfer to the black box model in question.

Section 3

min (| | δ | | p + c ⋅ f (x + δ)) , x + δ ∈ [0 , 1] n {\displaystyle \min(||\delta ||_{p}+c\cdot f(x+\delta)),x+\delta \in [0,1]^{n}}

Section 4

Evasion Evasion attacks consist of exploiting the imperfection of a trained model. For instance, spammers and hackers often attempt to evade detection by obfuscating the content of spam emails and malware. Samples are modified to evade detection; that is, to be classified as legitimate. This does not involve influence over the training data. A clear example of evasion is image-based spam in which the spam content is embedded within an attached image to evade textual analysis by anti-spam filters. Another example of evasion is given by spoofing attacks against biometric verification systems. Evasion attacks can be generally split into two different categories: black box attacks and white box attacks.

Section 5

Classifier influence: An attack can influence the classifier by disrupting the classification phase. This may be preceded by an exploration phase to identify vulnerabilities. The attacker's capabilities might be restricted by the presence of data manipulation constraints. Security violation: An attack can supply malicious data that gets classified as legitimate. Malicious data supplied during training can cause legitimate data to be rejected after training. Specificity: A targeted attack attempts to allow a specific intrusion/disruption. Alternatively, an indiscriminate attack creates general mayhem. This taxonomy has been extended into a more comprehensive threat model that allows explicit assumptions about the adversary's goal, knowledge of the attacked system, capability of manipulating the input data/system components, and on attack strategy. This taxonomy has further been extended to include dimensions for defense strategies against adversarial attacks.

Section 6

Feature robustness Researchers differentiate between features that can be easily manipulated and those that are more resistant to modification. For example, simple static attributes, such as header fields, may be altered by attackers, while structural features, such as control-flow graphs, are generally more stable but computationally expensive to extract.

Section 7

Fast gradient sign method One of the first proposed attacks for generating adversarial examples was proposed by Google researchers Ian J. Goodfellow, Jonathon Shlens, and Christian Szegedy. The attack was called fast gradient sign method (FGSM), and it consists of adding a linear amount of in-perceivable noise to the image and causing a model to incorrectly classify it. This noise is calculated by multiplying the sign of the gradient with respect to the image we want to perturb by a small constant epsilon. As epsilon increases, the model is more likely to be fooled, but the perturbations become easier to identify as well. Shown below is the equation to generate an adversarial example where x {\textstyle x} is the original image, ■ {\textstyle \epsilon } is a very small number, Δ x {\textstyle \Delta _{x}} is the gradient function, J {\textstyle J} is the loss function, θ {\textstyle \theta } is the model weights, and y {\textstyle y} is the true label.

Section 8

With this boundary function, the attack then follows an iterative algorithm to find adversarial examples x ′ {\textstyle x^{\prime }} for a given image x {\textstyle x} that satisfies the attack objectives.

Section 9

Adversarial natural language processing Adversarial attacks on speech recognition have been introduced for speech-to-text applications, in particular for Mozilla's implementation of DeepSpeech and Google's Speech Commands model. Various pre-processing techniques have been proposed to defend against such attacks , but these methods may not be resilient to attackers aware of those defenses.

Section 10

Labeling and ground truth challenges Malware labels are often unstable because different antivirus engines may classify the same sample in conflicting ways. Ceschin et al. note that families may be renamed or reorganized over time, causing further discrepancies in ground truth and reducing the reliability of benchmarks.

Section 11

Model extraction Model extraction involves an adversary probing a black box machine learning system in order to extract the data it was trained on. This can cause issues when either the training data or the model itself is sensitive and confidential. For example, model extraction could be used to extract a proprietary stock trading model which the adversary could then use for their own financial benefit. In the extreme case, model extraction can lead to model stealing, which corresponds to extracting a sufficient amount of data from the model to enable the complete reconstruction of the model. On the other hand, membership inference is a targeted model extraction attack, which infers the owner of a data point, often by leveraging the overfitting resulting from poor machine learning practices. Concerningly, this is sometimes achievable even without knowledge or access to a target model's parameters, raising security concerns for models trained on sensitive data, including but not limited to medical records and/or personally identifiable information. With the emergence of transfer learning and public accessibility of many state of the art machine learning models, tech companies are increasingly drawn to create models based on public ones, giving attackers freely accessible information to the structure and type of model being used.

Section 12

Simple black-box adversarial attacks Simple black-box adversarial attacks is a query-efficient way to attack black-box image classifiers. Take a random orthonormal basis v 1 , v 2 , … , v d {\displaystyle v_{1},v_{2},\dots ,v_{d}} in R d {\displaystyle \mathbb {R} ^{d}} . The authors suggested the discrete cosine transform of the standard basis (the pixels). For a correctly classified image x {\displaystyle x} , try x + ■ v 1 , x − ■ v 1 {\displaystyle x+\epsilon v_{1},x-\epsilon v_{1}} , and compare the amount of error in the classifier upon x + ■ v 1 , x , x − ■ v 1 {\displaystyle x+\epsilon v_{1},x,x-\epsilon v_{1}} . Pick the one that causes the largest amount of error.